

INTRODUCTION

1-1 WHY STUDY GEOMETRY?

There are at least four reasons why geometry is studied by so many students in the high schools of the United States and, in fact, throughout the world. The first of these reasons is that the logical beauty and precision of geometry has fascinated men since the time of the ancient Greeks. Geometry is one of the great achievements of the human mind, and for 2000 years men have regarded its study as a necessary part of the making of a truly educated person. The feeling has been that here is at least one thing whose truth is clear, because every statement can be demonstrated beyond a doubt. People also felt that because the reasoning in geometry had to be careful and accurate, the study of it would help one to be careful and accurate in other activities.* Because this great invention of the mind has value as an introduction to correct logical reasoning, geometry is a required subject for entrance to many colleges regardless of what one plans to study.

*The Dutch philosopher Benedictus de Spinoza (1632–1677) was so impressed by the clarity of geometrical reasoning that he arranged his reasoning on religion in a “geometric style.”

A second important reason for studying geometry is its practical importance. People in all kinds of occupations have occasional need for a bit of geometry, and in some fields the study of geometry is a most important step in professional training. Fields in which it is required are physics, chemistry, engineering, pure mathematics, statistics, some of the biological sciences, and, more recently, certain branches of economics and psychology.

Perhaps most students start the study of geometry because of its practical value or because they are told it is required. But a third reason is that after you have studied geometry you will have the knowledge that is necessary to give you a deeper insight into the wonderful details and complexities of our world, both the natural world and the constructions of mankind. It is this deeper understanding of the world which can be furthered by the study of geometry that makes its study important to all people, whether scientists or not. Nature abounds in geometric designs, and geometric knowledge gives the observer keener interest.

A fourth reason for studying geometry is that we live in what is often called a scientific age, and even if you yourself are not planning on a career in science it will be necessary for you to have some understanding and appreciation for the ways in which a scientist thinks. The study of geometry is a big step toward getting such understanding.

1-2 BRIEF HISTORY OF GEOMETRY

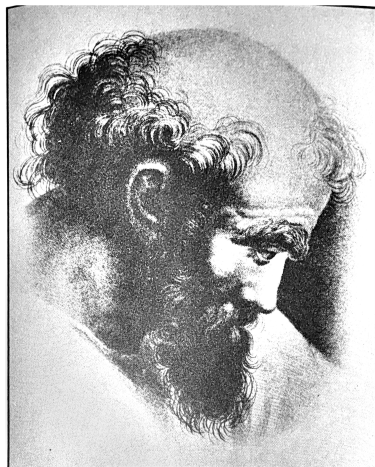
Knowledge of isolated geometric facts goes back before the beginning of recorded history. Certainly the early Egyptians and Babylonians (4000–3000 B.C.) knew many practically useful geometric relations. The construction of the pyramids alone required considerable practical geometry.

But it was the ancient Greeks who collected the known ge-

ometric facts, discovered new ones, and arranged them into a logically consistent system. (The word *geometry* comes from two Greek words, *ge* meaning “earth,” and *metrei* meaning “to measure,” showing that originally it was thought of as “earth measurement.”) This process of organization and discovery took centuries and occupied the minds of a long list of able men. Remember that the centuries just before, during, and after the period of greatest Greek political influence (4th and 5th centuries B.C.) were times of intense intellectual activity. The leading minds were interested in ideas of all kinds. Mathematics was but one of their interests.

Some of these early geometers deserve special mention either because of their mathematical eminence or because of their influence on later men of philosophy* or science. One of these early men was Thales of Miletos (6th century B.C.) who was considered one of the seven wise men of his day. He was the first Greek mathematician and the first Greek astronomer. Besides knowing some geometric facts, he probably knew proofs for many of them. The next important Greek for us was Pythagoras (late 6th century B.C.), who established a school, or brotherhood, in Croton, Sicily (a part of Greater Greece then). It is often hard to distinguish whether a discovery originated with Pythagoras or one of his followers. All ideas were communal property of the brotherhood. Besides the Pythagorean theorem concerning right triangles, he (or his school) made notable discoveries in music, astronomy, and arithmetic.

*The word *philosophy* comes from Greek words meaning “love of wisdom.” Look this up in your dictionary. The early Greek philosophers tried to understand the entire world. More on philosophy will be found in Chapter 16 of this text.



PYTHAGORAS

From a fresco by Raphael.
Courtesy of *Scripta Mathematica*.



THE DEATH OF ARCHIMEDES

From a mosaic found in the ancient city of Pompeii.
 Courtesy of *Scripta Mathematica*.

One Greek geometer in particular deserves mention here because his fame is so slight compared with his worth. This man is Eudoxus of Cnidus (4th century B.C.), one of the foremost mathematicians of all time. His fame rests upon his general theory of proportion, certain geometric constructions, and his “method of exhaustion.” He lived during the time of Plato.*

We find that the Greeks organized and extended geometry for their own mental enjoyment. Deductive geometry (that is, with proofs from simple assumptions) is truly a Greek invention. These men were not seeking practical applications at all; in fact, an educated Greek regarded applications of geometry as beneath his dignity. By the time of Plato (4th century B.C.), Greek scholars had a fairly well-organized body of geometry and perhaps even a geometric text written for members of Plato’s academy.

By far the best known of the ancient Greek geometers is Euclid, who wrote the *Elements* about 300 B.C. This work is the most successful textbook of all time and was used throughout the world until well into the present century. It consists of 13 “books,” the first six of which are concerned with plane geometry. The others deal with arithmetic and solid geometry. The *Elements* has not only been widely used as written (or as translated); it has also been the model for innumerable other books. Isaac Newton, the renowned English mathematician and physicist, wrote his great book, the *Principia*, in the “geometric style” even though this often tended to hide the path he took in making

*Plato and Aristotle are the two most renowned of the Greek philosophers. Although Eudoxus is less well known, he was a much better mathematician than either of these men. However, both Plato and Aristotle have had tremendous influence on later thought. It is said that above the entrance to Plato’s academy were inscribed the words, “Let no one ignorant of geometry enter here.”

his discoveries. Most of the geometry books used in American high schools today are somewhat altered and simplified versions of Euclid's *Elements*. It is astonishing, and a great tribute to Euclid, that this book has retained its value for over 2000 years.

Little is known of Euclid's life except that he probably learned his mathematics from pupils of Plato's academy and that he later founded his own school at Alexandria.[†] Undoubtedly he made full use of earlier writings, but the organization of the *Elements* is his own, as are many significant improvements.

One of the many amusing stories concerning Euclid is the following. Upon being asked by a student of what profit was the study of geometry, Euclid directed a slave to "give the man a coin that he may profit from what he learns."

Many able Greek mathematicians followed Euclid, among whom were Apollonius (3rd century B.C.) and Archimedes (3rd century B.C.). Archimedes, who had perhaps the greatest mind of ancient times, did fine work in mathematics, physics, and engineering. He was employed as a kind of technical consultant in warfare by the king of Syracuse and was killed by a soldier in the siege of that city in 212 B.C.

With the rise of the Roman Empire, Greece declined as a political power but retained its eminence as a center of learning and culture well into the Christian era. Rome added little (compared with Greece) to the arts, science, or literature. It is in the realm of practical government, warfare, and architecture that the Romans made their contributions. But when Rome fell to the barbarians from the north in the 5th century A.D., the "dark ages" began. During the dark ages, mathematical research was largely in the hands of the Arabs. (Our word *algebra* is of Arabic origin.) Serious scientific work in Europe was almost at a standstill until the

[†]The city of Alexandria was founded by Alexander the Great in 332 B.C. Alexander was a supporter of learning, as was his successor, Ptolemy I. It was Ptolemy who started the great library at Alexandria, and it was during his rule that Euclid opened a school there.

Renaissance, or revival of learning, in the 14th century.

Beginning with the 16th century, mathematical activity was primarily in fields other than classical geometry, particularly after the invention of calculus* by Newton and Leibniz in the 17th century. Nevertheless, significant studies in geometry were made which culminated in the discovery, in the 19th century by K. F. Gauss, N. Lobachevski, and J. Bolyai, independently, that geometries other than Euclid's are possible and just as true (that is, logically correct). It was during the 19th century that men became aware that there were certain assumptions made by Euclid which he did not actually state. Therefore Euclidean geometry was re-examined and made complete and precise according to modern standards. The first really complete treatment of Euclidean geometry was published by the German mathematician David Hilbert in 1899. Our book is based upon Hilbert's treatment (with some simplifications). Thus you will be studying geometry that is almost 2300 years old but dressed up in modern form.

Hilbert (who died in 1943 at the age of 81) had wide-ranging mathematical interests and made outstanding contributions to many fields. His influence on American mathematics has been considerable, because at the time when American mathematicians were beginning to achieve eminence, during the latter part of the 19th century, many of them studied under Hilbert at the University of Göttingen in Germany.

Suggested Discussion Topics

1. Discuss the four reasons others?
listed for studying geometry. Can you think of any
2. What do you know about

*Calculus is a powerful mathematical tool useful in solving certain kinds of problems. It is a subject required of most science majors in college.

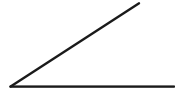
- Greek civilization other than what has been mentioned in this chapter?
3. What were some of the major scientific developments of the 19th century? In what countries did they occur?
 4. Look up and report on contributions of the Arabs to mathematics.
 5. Look up Euclid in an encyclopedia.
 6. Look up Pythagoras in an encyclopedia.
 7. Look up Archimedes in *The World of Mathematics*, by J. R. Newman.

1-3 FAMILIAR GEOMETRIC OBJECTS

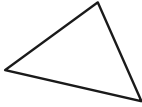
In your previous study of arithmetic and algebra you have met many “geometric objects,” such as the ones shown in Fig. 1-1, where the examples are all *plane* figures. Some *solid* figures that you may have encountered are pictured in Fig. 1-2.



Line segment

Ray from O 

Angle



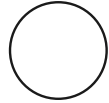
Triangle



Square



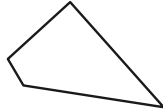
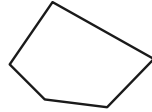
Parallelogram



Circle

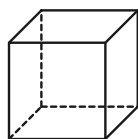


Trapezoid

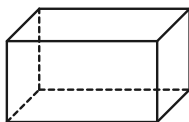
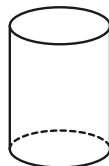
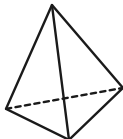
Quadrilateral, or
4-sided polygonPentagon, or
5-sided polygon

Rectangle

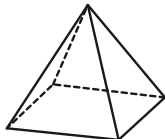
FIGURE 1-1



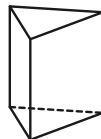
Cube

Rectangular
parallelepiped or boxCircular
cylinder

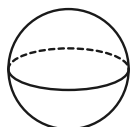
Tetrahedron



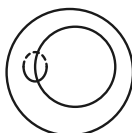
Pyramid



Prism



Sphere

Torus, or doughnut,
or anchor ring

Cone

FIGURE 1-2

All these shapes are easily recognized as familiar; you would have no difficulty in recognizing one of them on sight. However, you may have noticed that we have not *defined* the shapes, but only drawn them. Part of our job later in this book will be to give accurate definitions. As you study geometry, it will become more and more clear that we cannot prove anything without a clear definition. But in this chapter, we shall most often use drawings to introduce the things you will study later.

We have used the words “geometric object” or “geometric figure,” but have only given examples by drawings on a page of the book. A precise definition is the following:

Definition 1-1. A geometric figure is any *set of points*.

Some explanation of the words “set of points” is necessary. A set of points is just a “bunch” of points (or “collection” of points, or “aggregate” of points). This definition allows some rather strange sets of points to be called geometric figures, for example those in Fig. 1–3. However, we usually deal with simpler sets of points than these, namely, with lines, line segments, triangles, circles, etc. A word of caution is necessary regarding our use of the word “line.” By “line” we mean a *straight line* indefinite in extent in both directions. We shall see in Chapter 3 exactly how to state what we need to know about lines in order to prove other things about them.

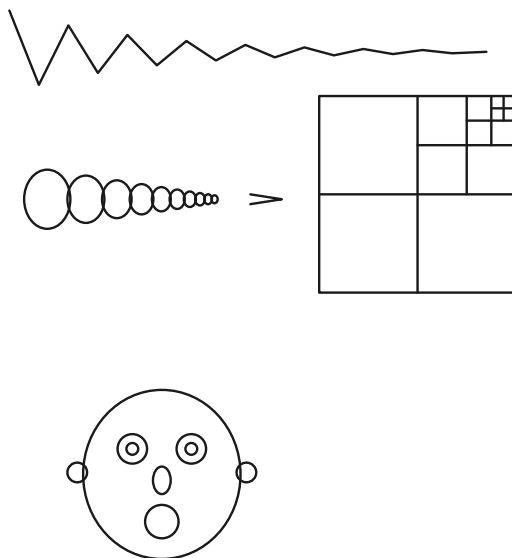


FIGURE 1–3

Exercise Group 1–1

1. Describe in your own words what is meant by a *set* of points.
2. Invent some geometric figures of your own and then describe these sets of points.
3. Mark a point on a sheet of paper and consider the set of all points on your paper such that each point in the set is two inches from this first point. Describe this set of points in another way.
4. Some of the shapes drawn in Fig. 1–3 are intended to represent infinitely many line segments or circles. Which are they? Can you invent other figures with infinitely many segments or circles?

1–4 WHAT ARE POINTS?

Up to now we have been talking about points just as if we knew what they are. But no one has ever seen a point. The smallest dot you can make on a sheet of paper is certainly not a point. The dot would look huge to a microbe (if it could see), and the microbe, which can be seen under a microscope, is very large compared with the molecules of chemical substances of which it is composed. What, then, is a point? The answer is that points are an *invention* of the human mind. We *imagine* that there *ought to be* points, which themselves cannot be split up into smaller pieces. A given “hunk” or portion of space can be split into two hunks. And each of these can be split in two, and so on indefinitely. A point is what *should* be left when our hunk has no extension in space or, to use other words, when it is “infinitely small.” Euclid, in fact, gave something like that as a definition of a point. But such a “definition” is not really a help in geometry because from it we cannot derive anything more. All it does is make us feel more or less happy about the existence of the things we *think* we are talking about. The modern point of view (and really Euclid’s too) is that points are undefined. The things that we do define are certain relations between points or sets of points.

It is essentially the same regarding lines and planes. No one

has ever seen a line or a plane. A thin ray of light or a thin line on the page is only an approximation to what we imagine that a line should be like. And a very flat surface, like a mirror, is but an approximation to what we imagine a plane to be. Therefore right at the beginning of geometry we have to make a strong effort of the imagination. We must imagine a large set of things, called points, which we call a plane. Certain parts of this plane, that is, certain special sets of points in the plane, are called straight lines, or just lines. The study of geometry begins when we tell *how these points and lines are related to each other.*

Exercise Group 1–2

1. Draw two intersecting lines on the blackboard. You will then have some chalk hanging on the board. How many reasons can you give to explain that what you have drawn are not lines? Does the chalk on both your drawn lines correspond to what you think of as a point?
2. Mark two dots on the blackboard. Are they points? Is a “blob” of chalk a point? How many lines do you feel there should be which lie on two given points? If points were blobs of chalk, could you draw more than one line through them?
3. When you draw a figure to illustrate a geometric idea, what is the relation of the drawn figure to the geometric idea you have in mind?
4. Is it reasonable to expect a drawing to give a proof of something in geometry?
5. If an object had length or width, would you call it a point? Can you see points and lines?
6. Can you describe what you mean by straightness of a line?

1–5 TOOLS YOU WILL NEED

Even though points and lines are creations of the mind it is necessary for us to draw pictures of them in order to see what is “go-

ing on.” Thus you will need a *ruler* to draw pictures of straight lines. A ruler without a marked scale on it is called a *straightedge*, and for the constructions we shall make this is all that is absolutely necessary. But it is very convenient to be able to *measure* the length of a line segment, and for this reason an ordinary foot ruler, or a six-inch one, will do. However, we shall not actually deal with measurement* of lengths until Chapter 4.

Angles will be discussed more fully later, and we shall consider measurement of angles in Chapter 12. But you are already familiar with a device, called the *protractor*, for measuring pictures of angles drawn on paper or on the blackboard.

You will also need a device, called a *compass*, for drawing pictures of circles. Undoubtedly you are familiar with its use.

With these simple tools you can make a number of “constructions.” These constructions will seem to be obviously correct. However, we will prove later in the book that they actually are correct.

You have learned in your previous mathematics how to use these tools. The exercises below are for review of definitions and practice in the use of ruler, compass, and protractor. However, throughout this course your best tool will be your ability to think straight.

Exercise Group 1–3

1. Draw a picture of a line, using your straightedge. Now if you sharpened your pencil *very sharp*, could you draw a picture of a line inside the first one? Was your drawing really what you intended a line to be?
2. Draw a picture of a circle with your compass. How many dif-

*It is interesting and surprising that Euclid in his *Elements* did not discuss measurement, that is, the process of getting a *number* for the distance between two points.

ferences can you find between your drawing and an imaginary perfect circle?

3. Draw a picture of an angle. Measure it with your protractor. Have a friend measure the angle. Do you get *exactly* the same answers? How accurate do you think your measurement is?
4. How long are the sides of an angle? When you draw a picture of an angle, you cannot draw the sides infinitely long because your straightedge and paper are limited in size.
5. Draw a picture of a ray from a point A. Draw another ray from A. What we call the *angle* consists of the points on the two rays. The points between the rays form the *interior* of the angle. The other points of the plane form the *exterior* of the angle. (Fig. 1-4.)

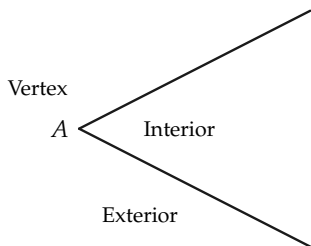


FIGURE 1-4

6. A *right angle* has a measure of

90° . Construct a right angle with your protractor.

7. Draw an angle of 37° , of 120° , of 85° .
8. An angle is *bisected* if a ray from the vertex splits the angle into two angles of the same size. In Fig. 1-5, the ray AC bisects $\angle BAD$ because $\angle CAB$ and $\angle DAC$ are the same size. The ray AC is the bisector of the angle. Draw an angle of 45° and bisect it, using your protractor.

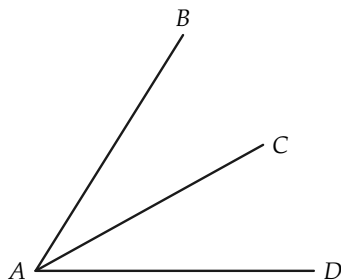


FIGURE 1-5

9. Draw a picture of a triangle. Measure each of the angles of the triangle. What is their sum? Repeat for several different triangles. Can you guess something that is true for all triangles?
10. As in Fig. 1-6, an *acute angle* is an angle of less than 90° ; an *obtuse angle* is an angle of more than 90° . Draw two acute

angles and two obtuse angles with your protractor.

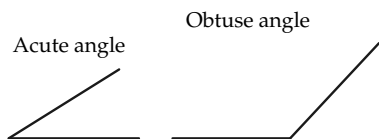


FIGURE 1-6

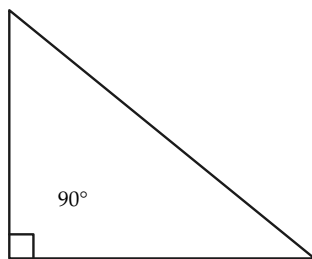


FIGURE 1-7

11. Draw a triangle and measure the sides and the angles. Which side is opposite the largest angle? opposite the smallest angle? Draw another triangle and repeat the measurements. Can you guess something which now seems to be true for all triangles?
12. A triangle that has one right angle is called a *right triangle*. Draw a picture of a right triangle, as in Fig. 1-7. Measure the two acute angles of the triangle. What is their sum? Draw another right triangle and make the same measurements. Is the sum of the two angles the same as before?

13. One angle is said to be *complementary* to another if their sum is 90° . Construct two such angles with your protractor. In Fig. 1-8, if $\angle AOB$ is a right angle, what can you say about $\angle COA$? about $\angle COB$?

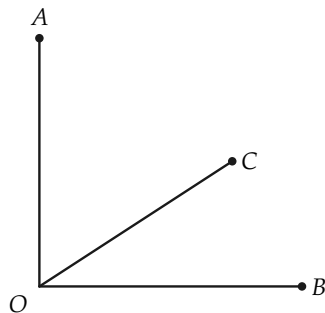


FIGURE 1-8

14. One angle is said to be *supplementary* to another if their sum is 180° . Construct two such angles with your protractor. What can you say about $\angle AOB$ and about $\angle BOC$ in Fig. 1-9?

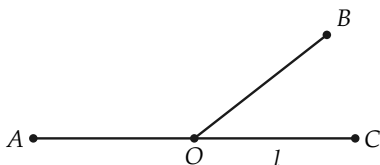


FIGURE 1-9

15. Two distinct rays OA and OB which lie on one straight line form a *straight angle* (Fig. 1-10). What is the measure of a straight angle?



FIGURE 1-10

16. A *quadrilateral* (Fig. 1-11) is a figure with four sides, just as a triangle is a figure with three sides. Draw a picture of a quadrilateral and measure its angles. What is their sum?

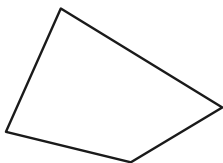


FIGURE 1-11

17. Add the following:

- (a) $12^\circ 10' 25''$, $31^\circ 48' 30''$, and $43^\circ 50' 40''$.
 (b) $17^\circ 18' 19''$, $90^\circ 9' 9''$, and $41^\circ 48' 48''$.

- (c) $121^\circ 18' 17''$, $40^\circ 40' 40''$, and $18^\circ 1' 3''$.

- (d) $66^\circ 55' 44''$, $55^\circ 44' 33''$, and $44^\circ 33' 22''$.

18. Subtract:

- (a) $41^\circ 15' 13''$ from $63^\circ 28' 44''$.

- (b) $41^\circ 15' 13''$ from $50^\circ 18' 7''$.

- (c) $61^\circ 28' 37''$ from $80^\circ 18' 23''$.

19. What is the complement of an angle of

- (a) $51^\circ 17' 18''$?

- (b) $42^\circ 38' 29''$?

- (c) A° ?

20. What is the supplement of an angle of

- (a) $51^\circ 17' 18''$?

- (b) $138^\circ 4' 5''$?

- (c) A° ?

21. If an angle is twice as large as its complement, what is its degree measure? [Hint: Set up an equation to solve.]

22. Two angles have a sum of $56^\circ 28'$ and a difference of $8^\circ 46'$. Find the angles.

23. Find the complement and supplement of an angle of $56^\circ 28' 31''$. What is the difference between the supplement and complement? Can you guess something that is true for all acute angles?

24. What kind of angle is the supplement of an acute angle? of an obtuse angle? Does an obtuse angle have a complement?

1-6 RULER AND COMPASS

During their study of geometry, the ancient Greeks found that many interesting figures could be drawn with a straightedge and compass alone. Drawings needing only these tools are called *ruler and compass constructions*. The Greeks tried to solve all construction problems using only these tools, but they encountered some that could not be done this way.

The exercises below involve some simple ruler and compass constructions that the Greeks knew. Exercise 2, in fact, is the first proposition in Euclid's *Elements*. We shall return to these constructions again in Chapter 7, where we shall *prove* that some of them are correct.

Exercise Group 1-4

1. A triangle having two sides of equal length is called an *isosceles triangle*. Draw a picture of an isosceles triangle. Measure the angles opposite the equal sides. Draw several isosceles triangles, each with the equal sides three inches long.
2. A triangle with all sides the same length is called an *equilateral triangle* (Fig. 1-12). Draw a picture of an equilateral triangle. This is easy to do with your compass and straightedge. What is the degree measure of each angle in an equilateral triangle? A triangle with all angles equal is called an *equiangular triangle*. Are equilateral triangles equiangular? Are equiangular triangles equilateral?

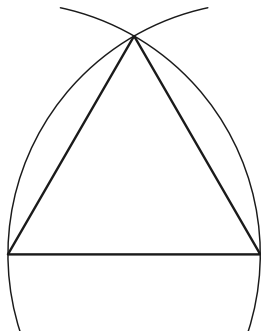


FIGURE 1-12

3. Draw an angle like that in Fig. 1-13. Use your compass to copy $\angle AOB$ as $\angle A'O'B'$ in the figure. Measure the two angles with your protractor to see if they are the same size.

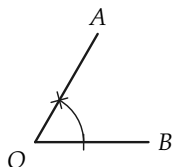
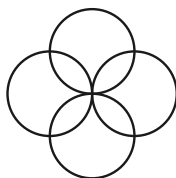


FIGURE 1-13

4. Draw the figures shown in Fig. 1-14.



5. Use your straightedge and compass to draw and bisect an angle (Fig. 1-15). Check your work by measurement with a protractor. State in your own words why you think the construction is correct.

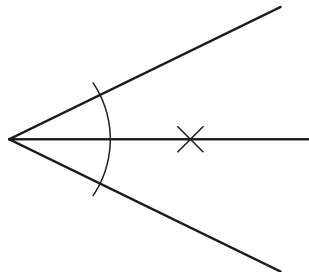


FIGURE 1-15

6. Angles which have a common vertex and a common side and have no interior points in common are said to be *adjacent*. Thus, in Fig. 1-16, $\angle AOB$ is adjacent to $\angle BOC$. Is $\angle AOC$ adjacent to $\angle BOC$? Are the angles in Fig. 1-17 adjacent to one another?

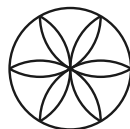
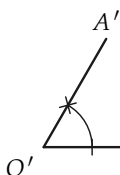


FIGURE 1-14

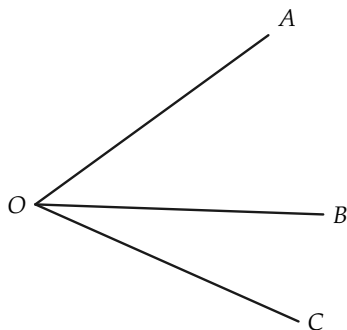


FIGURE 1-16

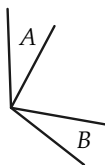
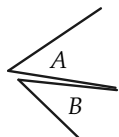


FIGURE 1-17

7. Draw a picture of a segment. From some point A on a line, construct with your compass a segment AB as long as the segment you first drew. (See Fig. 1-18.)

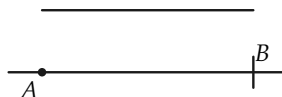


FIGURE 1-18

8. Two lines are *perpendicular* if they intersect at right angles. Another way of defining perpendicular would be to say that all the adjacent angles formed by the intersecting lines are the same size. To indicate that l and m are perpendicular, we write $l \perp m$.

A line l is called the *perpendicular bisector* of the segment AB if the line l is perpendicular to AB and intersects the segment AB in a point C such that AC and CB have equal lengths. The point C is called the *midpoint* of AB . (See Fig. 1-19.)

Draw a line segment and construct its perpendicular bisector with straightedge and compass. Check by measurement. State why you think the construction is correct.

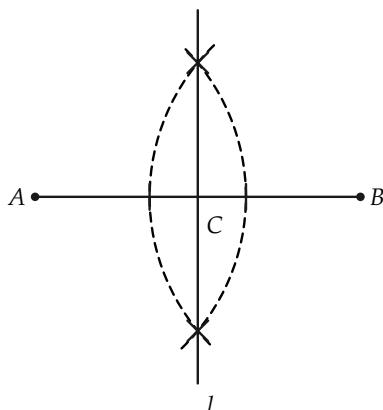


FIGURE 1-19

9. Draw a line l and mark a point O on it. With straightedge and compass, construct a line $m \perp l$ and passing through O . (See Fig. 1-20.)

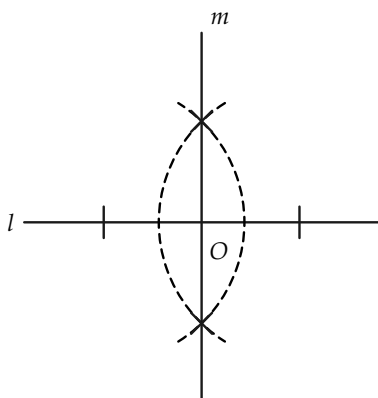


FIGURE 1-20

10. Draw a line l and mark a point P not on the line.

With your straightedge and compass, construct a line m through P perpendicular to l . Check by measurement with your protractor. State why you think the construction is correct. (See Fig. 1-21.)

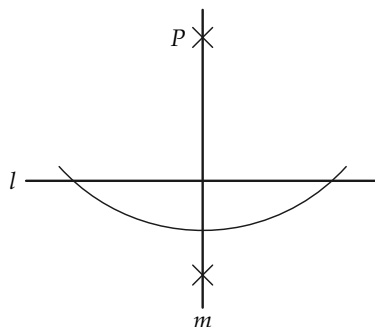


FIGURE 1-21

11. A line segment drawn from a vertex of a triangle to the midpoint of the opposite side is called a *median* of the triangle. A triangle has three medians that meet in one point O (see Fig. 1-22). We shall prove this later.
- Draw a triangle and its medians. Can you describe where the medians intersect? Try to find how to describe O by measurement. Make this construction for several triangles.

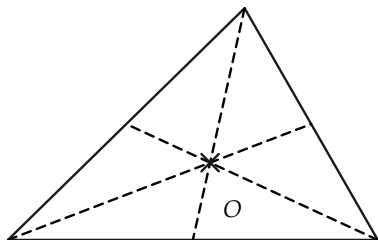


FIGURE 1-22

12. Cut a triangle out of cardboard. Find where the medians intersect. If your work is accurate you will find that the triangle will balance on your finger at the place where the medians intersect (Fig. 1-23).

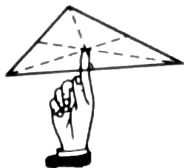
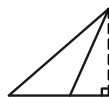


FIGURE 1-23

13. An *altitude* of a triangle is a line segment drawn from a vertex perpendicular to the line *containing* the opposite side [see Fig. 1-24, (a) and (b)]. A triangle has three altitudes. The lines containing the altitudes of a triangle always intersect in one point [see Fig. 1-24(c)]. Draw several triangles and their altitudes.



(a)



(b)

FIGURE 1-24

14. Draw a triangle and bisect each of the angles of the triangle. The three bisectors will meet in one point (Fig. 1-25).

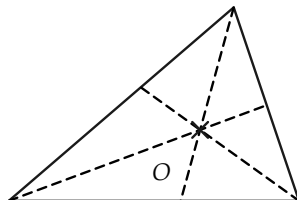


FIGURE 1-25

15. Two lines, l and m , are said to be *parallel* if they do not intersect. We indicate " l is parallel to m " by writing $l \parallel m$. (Remember that lines are infinitely long.) If another line n intersects the parallel lines, then the angles 1 and 2 in Fig. 1-26 are the same size. Conversely, if angles 1 and 2 are the same size, then the lines are parallel. As in Fig. 1-27, draw a line l and a line n intersecting it. Choose a point P on n but not on l . At P , copy angle 2 in order to get a line parallel to

1. Make this construction with your protractor and then repeat it with your straightedge and compass.

We shall prove later that this construction with straightedge and compass is correct.

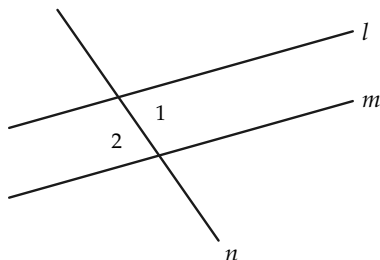


FIGURE 1-26

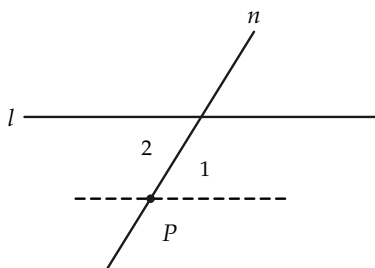


FIGURE 1-27

16. Draw a line l and mark a point A not on l . Construct a line

through the point A parallel to l .

17. Draw a right triangle (Fig. 1-28). The side opposite the right angle is called the *hypotenuse*. The other two sides are called the *legs* of the triangle. Measure the lengths of the sides in your right triangle. If a , b , and c are the lengths of the sides, where c is the hypotenuse, you will find that $a^2 + b^2 = c^2$. This formula is called the *Pythagorean theorem*. It was discovered (probably) by the Greek Pythagoras in the 6th century B.C. We will prove this important theorem later in the book. Test the truth of the Pythagorean theorem by measuring several different right triangles.

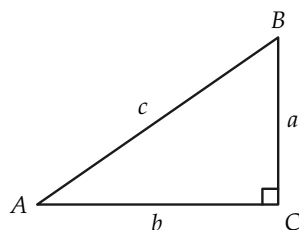


FIGURE 1-28

REVIEW OF CHAPTER 1

1. Make sure that you are familiar with all the following terms:

line segment	perpendicular
angle	right angle
angle bisector	equilateral
triangle	equiangular
degree, minute, second	isosceles
midpoint	median
complementary	altitude
supplementary	hypotenuse
straight angle	diameter
obtuse angle	radius
acute angle	adjacent angles
ray	intersect
set	contain

2. Make sure that you can do the following straightedge and compass constructions:

- (a) Copy a segment.
- (b) Draw the bisector of a segment.
- (c) Draw a line $\perp$ to a given line from a point not on the given line.
- (d) Draw a line $\perp$ to a given line from a point on a given line.
- (e) Draw a line $\parallel$ to a given line, through a point not on the line.
- (f) Bisect an angle.
- (g) Copy an angle.

Brief Review Tests

Decide whether the following statements are true or false. If in doubt, look up the answer. Do not guess.

I. History

1. Euclid lived before Pythagoras.
2. Euclid collected, organized, and added to the geometry of his day.
3. Euclid's geometry was incomplete as he wrote it. It was completed in the 19th century.
4. Archimedes lived at the same time as Euclid.
5. Thales lived before Euclid.
6. Greece was superior to Rome in all ways.
7. Rome was a great center for the study of geometry.

II. *Geometry*

1. A straightedge is used for measurement.
2. A line may have finite length.
3. A straight angle contains 90° .
4. If angles have a common vertex and a common side, they are adjacent.
5. Complementary angles have a sum of 180° .
6. In a right triangle the side opposite the right angle is called the hypotenuse.
7. The altitude of a triangle is its distance from the center of the earth.
8. If two lines intersect they are parallel.

ARITHMETIC AND ALGEBRA REFRESHER

1. Carry out the arithmetic operations indicated:

- | | |
|-------------------------------------|-------------------------------------|
| (a) $[(12)(8)] \div 6$ | (i) $\frac{1}{2} \cdot \frac{1}{3}$ |
| (b) $2992 \div 68$ | (j) $\frac{1}{2} \div \frac{1}{3}$ |
| (c) $326 + (-257)$ | (k) $(-24) \div (-6/5)$ |
| (d) $\frac{1}{2} + \frac{1}{4}$ | (l) $[(-6.1) + (-8.5)] - 2.7$ |
| (e) $\frac{1}{3} \div \frac{11}{3}$ | (m) $3/7 \cdot 35/24$ |
| (f) $56789 \div 357$ | (n) $2.5679/0.1372$ |
| (g) $\frac{2}{3} - \frac{2}{5}$ | (o) $[3.4 - 8.5] + (-2.7)$ |
| (h) $3.762 \div 5.7$ | (p) $(5 + \frac{1}{2})/33$ |

2. Write the answers to the following. If you take more than a minute and miss any part of this problem, you need more review.

- | | | |
|----------------------|----------------------------|-----------------------------------|
| (a) $2 \times 2 =$ | (d) $2 \div 0.2 =$ | (f) $2 \times \frac{1}{2} =$ |
| (b) $2 \div 2 =$ | | (g) $\frac{1}{2} + \frac{1}{3} =$ |
| (c) $2 \times 0.2 =$ | (e) $2 \div \frac{1}{2} =$ | |

3. Write formulas for the following. Be sure that you tell what each letter stands for. For example: If A is the area of a rectangle l its length and w its width, then $A = lw$.

- (a) The volume of a box.
- (b) The area of a circle.
- (c) The circumference of a circle.
- (d) The area of a triangle.
- (e) The product of a number z , by one more than the number.
- (f) Twice a number plus 5 is equal to 13.
- (g) One number is 4 times another.

4. Find x if

- | | |
|------------------------|------------------------------------|
| (a) $2x - 5 = 3$ | (f) $\frac{1}{3} \cdot (12x) = 44$ |
| (b) $3x - 5 = x + 7$ | (g) $2x/3 + 1 = 5$ |
| (c) $2x = 3$ | (h) $4x + 5x - x = 96$ |
| (d) $ax = b, a \neq 0$ | (i) $x \div 6 = 5 \div 24$ |
| (e) $x^2 = 4$ | (j) $6.4 = 2x - 0.4$ |

5. If x and y are numbers, what do we call xy ? $x + y$?

6. Write equations for the following statements:

- (a) The sum of two numbers is 17 and their difference is 7.
- (b) A number z is equal to the sum of the product of x and y , and the product of a and b .
- (c) The number π is equal to the quotient of the circumference of a circle divided by its diameter.
- (d) The volume of a sphere is equal to $\frac{4}{3}$ of the product of π and the cube of the radius.
- (e) Einstein's famous formula for the equivalence of mass and energy is as follows: If E is the energy of a mass m , and c is the velocity of light, in suitable units, then E is equal to the product of the mass by the square of the velocity of light. Write this formula.